



VP AIR TEST NEET (2025-26)

PART TEST - 05

DROPPER

DURATION: 180 Minutes

Date: 16-11-2025

M. MARKS: 720

ANSWER KEY

1. (2)	46. (1)	91. (2)	136. (3)
2. (3)	47. (4)	92. (4)	137. (2)
3. (2)	48. (1)	93. (1)	138. (3)
4. (3)	49. (4)	94. (2)	139. (1)
5. (3)	50. (2)	95. (1)	140. (1)
6. (4)	51. (3)	96. (1)	141. (1)
7. (2)	52. (3)	97. (4)	142. (4)
8. (2)	53. (2)	98. (1)	143. (2)
9. (3)	54. (3)	99. (2)	144. (2)
10. (2)	55. (3)	100. (2)	145. (3)
11. (3)	56. (1)	101. (1)	146. (4)
12. (2)	57. (4)	102. (1)	147. (2)
13. (1)	58. (3)	103. (3)	148. (3)
14. (1)	59. (1)	104. (1)	149. (4)
15. (2)	60. (3)	105. (4)	150. (3)
16. (2)	61. (2)	106. (2)	151. (2)
17. (4)	62. (2)	107. (4)	152. (3)
18. (2)	63. (4)	108. (4)	153. (3)
19. (2)	64. (1)	109. (4)	154. (4)
20. (1)	65. (1)	110. (2)	155. (3)
21. (2)	66. (1)	111. (3)	156. (4)
22. (2)	67. (2)	112. (3)	157. (3)
23. (3)	68. (2)	113. (4)	158. (3)
24. (4)	69. (4)	114. (3)	159. (2)
25. (2)	70. (1)	115. (2)	160. (4)
26. (1)	71. (2)	116. (3)	161. (1)
27. (3)	72. (1)	117. (2)	162. (2)
28. (3)	73. (2)	118. (3)	163. (1)
29. (1)	74. (2)	119. (3)	164. (1)
30. (1)	75. (2)	120. (4)	165. (4)
31. (1)	76. (2)	121. (3)	166. (2)
32. (4)	77. (3)	122. (3)	167. (4)
33. (4)	78. (2)	123. (1)	168. (2)
34. (4)	79. (3)	124. (3)	169. (1)
35. (2)	80. (3)	125. (1)	170. (2)
36. (1)	81. (2)	126. (3)	171. (1)
37. (2)	82. (3)	127. (3)	172. (3)
38. (3)	83. (1)	128. (3)	173. (2)
39. (4)	84. (1)	129. (1)	174. (3)
40. (4)	85. (3)	130. (3)	175. (1)
41. (2)	86. (2)	131. (2)	176. (3)
42. (4)	87. (2)	132. (2)	177. (3)
43. (2)	88. (1)	133. (4)	178. (1)
44. (4)	89. (3)	134. (1)	179. (3)
45. (1)	90. (2)	135. (2)	180. (2)

1. (2)

First law (sign convention: work by the gas is +W):

$$\Delta U = Q - W$$

$$Q = -200 \text{ J}, W = +50 \text{ J}$$

$$\Rightarrow \Delta U = Q - W = -200 - 50 = -250 \text{ J}$$

(NCERT Ed. 2025-26, 11th, Page No. 230)

2. (3)

Mean free path:

$$\lambda = \frac{1}{\sqrt{2}n\sigma}, \quad \sigma = \pi d^2$$

$$\text{so } \lambda \propto \frac{1}{d^2} = d^{-2}. \text{ Hence } n = -2.$$

(NCERT Ed. 2025-26, 11th, Page No. 255)

3. (2)

$$\eta_1 = \frac{1}{3} \Rightarrow T_c / T_h = \frac{2}{3} \Rightarrow T_c = \frac{2}{3} T_h.$$

With new sink $T_c = 60$,

$$\frac{1}{2} = 1 - \frac{T_c - 60}{T_h} = 1 - \left(\frac{\frac{2}{3} T_h - 60}{T_h} \right) = \frac{1}{3} + \frac{60}{T_h}$$

$$\Rightarrow \frac{1}{6} = \frac{60}{T_h} \Rightarrow T_h = 360 \text{ K}.$$

(NCERT Ed. 2025-26, 11th, Page No. 238)

4. (3)

$$\Delta V = 4 \text{ L} = 4 \times 10^{-3} \text{ m}^3, \quad P = 2 \times 10^5 \text{ Pa}.$$

$$P \Delta V = (2 \times 10^5)(4 \times 10^{-3}) = 800 \text{ J}.$$

For monoatomic isobaric:

$$Q = \frac{5}{2} P \Delta V = \frac{5}{2} \times 800 = 2000 \text{ J}.$$

(NCERT Ed. 2025-26, 11th, Page No. 253)

5. (3)

In compression, external work is done on the gas ($W < 0$). With $Q = 0, \Delta U = -W > 0 \Rightarrow \Delta T > 0$.

A is true. The reason is a correct statement and directly explains A.

(NCERT Ed. 2025-26, 11th, Page No. 235)

6. (4)

$$\frac{Q}{t} = \frac{KA \Delta T}{\Delta x}$$

$$[K] = \frac{[Q \Delta x]}{[A t \Delta T]} = \frac{[ML^2 T^{-2} L]}{[L^2 T \theta]} = [MLT^{-3} \theta^{-1}]$$

(NCERT Ed. 2025-26, 11th, Page No. 214)

7. (2)

Start from the adiabatic law and ideal-gas equation:

$$PV^\gamma = \text{const}, PV = nRT \rightarrow V = \frac{nRT}{P}$$

Substitute V into PV^γ :

$$P \left(\frac{nRT}{P} \right)^\gamma = (nR)^\gamma T^\gamma P^{1-\gamma} = \text{const}.$$

Hence,

$$P^{1-\gamma} T^\gamma = \text{constant}.$$

General form is $P^{1-\gamma} T^\gamma = \text{const}.$ For $\gamma = 5/3: 1 - \gamma = -2/3$.

$$\text{So } P^{-2/3} T^{5/3} = \text{const}.$$

(NCERT Ed. 2025-26, 11th, Page No. 235)

8. (2)

For r.m.s. speed: $v_{rms} \propto \sqrt{\frac{T}{M}}$ (M: molar mass).

$$v_{rms} \propto \sqrt{T} \Rightarrow \frac{T_2}{T_1} = \left(\frac{v_2}{v_1} \right)^2 = (1.5)^2 = 2.25.$$

(NCERT Ed. 2025-26, 11th, Page No. 250)

9. (3)

$$R_A = 2\sqrt{h(H-h)} = 2\sqrt{\left(\frac{H}{6}\right)\left(\frac{5H}{6}\right)}$$

$$R_B = 2\sqrt{\left(\frac{H}{2}\right)\left(\frac{H}{2}\right)}$$

$$x = \frac{R_A}{R_B} = \sqrt{\frac{5}{36}} \times 4 = \frac{\sqrt{5}}{3}$$

(NCERT Ed. 2025-26, 11th, Page No. 188)

10. (2)

$$U = 2\left(\frac{5}{2}R\right)T + 1\left(\frac{3}{2}R\right)T = 5RT + 1.5RT = \frac{13}{2}RT$$

(NCERT Ed. 2025-26, 11th, Page No. 253)

11. (3)

At constant volume:

$$C_{V,mix} = 2\left(\frac{3}{2}R\right) + 1\left(\frac{5}{2}R\right) = 3R + 2.5R = \frac{11}{2}R.$$

Total moles $n = 3$.

$$\text{Hence } C_{P,mix} = C_{V,mix} + nR = \frac{11}{2}R + 3R = \frac{17}{2}R.$$

$$\gamma = \frac{C_{P,mix}}{C_{V,mix}} = \frac{17/2}{11/2} = \frac{17}{11}$$

(NCERT Ed. 2025-26, 11th, Page No. 253)

12. (2)

Pressure on the concave side is higher than the convex side.

(NCERT Ed. 2025-26, 11th, Page No. 196)

13. (1)

Moment of inertia of rod $I = KL^2$

$$\frac{\Delta I}{I} = \frac{2\Delta L}{L} = \frac{2L\alpha\Delta T}{L}$$

$$\Rightarrow \frac{\Delta I}{I} = 2\alpha\Delta T$$

(NCERT Ed. 2025-26, 11th, Page No. 205)

14. (1)

Both statements are correct.

Continuity equation $\rightarrow$ conservation of mass.

Bernoulli's equation applies (in its standard form) to steady, non-viscous, incompressible flow along a streamline.

(NCERT Ed. 2025-26, 11th, Page No. 188)

15. (2)

Rate of heat emission

$$\frac{dQ}{dt} = e\sigma AT^4$$

$$\left(\frac{dQ}{dt}\right)_1 = \frac{e_1}{e_2} \times \frac{A_1}{A_2} \times \frac{T_1^4}{T_2^4}$$

$$= \frac{1}{4} \times \frac{2^4}{1} = 4:1$$

(NCERT Ed. 2025-26, 11th, Page No. 219)

16. (2)

Isothermal $\rightarrow \Delta U = 0$.

Expansion $\Rightarrow V_2 > V_1 \Rightarrow W = nRT \ln(V_2 / V_1) > 0$.

(NCERT Ed. 2025-26, 11th, Page No. 234)

17. (4)

$$Q = mc_{ice}\Delta T + mL + mc_w\Delta T$$

$$= (10 \times 0.5 \times 10) + (10 \times 80) + (10 \times 1 \times 20)$$

$$= 50 + 800 + 200 = 1050 \text{ cal}$$

(NCERT Ed. 2025-26, 11th, Page No. 214)

18. (2)

Assuming the temperature of surrounding be $t^\circ\text{C}$

Case-I

$$\frac{(60-50)}{4} = k(55-t) \dots(1)$$

Case-II

$$\frac{(40-30)}{8} = k(35-t) \dots(2)$$

Divide (1) and (2)

$$\frac{10/4}{10/8} = \frac{(55-t)}{(35-t)}$$

$$\Rightarrow 2(35-t) = 55-t$$

$$\Rightarrow 70 - 2t = 55 - t$$

$$\Rightarrow t = 15^\circ\text{C}$$

(NCERT Ed. 2025-26, 11th, Page No. 220)

19. (2)

$$R = n^{1/3}r.$$

$$\Delta E = \sigma[n(4\pi r^2) - 4\pi R^2] = 4\pi\sigma r^2(n - n^{2/3}).$$

(NCERT Ed. 2025-26, 11th, Page No. 194)

20. (1)

Atmospheric head $h_a = 10 \text{ m}$.

Bottom: $h_b = h_a + H$.

At 1 m depth: $h_1 = h_a + 1 = 11 \text{ m}$.

Given $h_b = 6h_1 \Rightarrow h_a + H = 6 \times 11 = 66$

$$\Rightarrow H = 56 \text{ m}.$$

(NCERT Ed. 2025-26, 11th, Page No. 184)

21. (2)

Applying equation of continuity

$$2Av_1 = 2Av_2 + Av_3$$

$$2 \times 4 = 4 + v_3$$

$$v_3 = 4 \text{ m/s}$$

(NCERT Ed. 2025-26, 11th, Page No. 187)

22. (2)

With the decrease in the temperature, the most probable speed decrease but corresponding number of molecules having most probable speed increases.

(NCERT Ed. 2025-26, 11th, Page No. 250)

23. (3)

If potential energy is negligible, the gas's energy is only kinetic; average kinetic energy is $\propto T$, so internal energy depends only on T.

(NCERT Ed. 2025-26, 11th, Page No. 250)

24. (4)

Faulty scale: $t_f = 0^\circ\text{C} \rightarrow 5^\circ\text{C}$,

$$100^\circ\text{C} \rightarrow 99^\circ\text{C} \Rightarrow t_f = 0.94T + 5.$$

At $(t_f = 52)$, $52 = 0.94T + 5 \Rightarrow T = 50^\circ\text{C}$.

$$\text{Convert: } F = \frac{9}{5}T + 32 = \frac{9}{5} \cdot 50 + 32 = 122^\circ\text{F}.$$

(NCERT Ed. 2025-26, 11th, Page No. 203)

25. (2)

For air bubble: $\Delta P \propto 1/R$.

$$\frac{\Delta P_1}{\Delta P_2} = \frac{R_2}{R_1} = \frac{3}{2}.$$

(NCERT Ed. 2025-26, 11th, Page No. 196)

26. (1)

$$\lambda_m T = \text{constant}$$

$$\Rightarrow \frac{(\lambda_m)_1}{(\lambda_m)_2} = \frac{T_2}{T_1} = \frac{(3227 + 273)}{(2227 + 273)}$$

$$\Rightarrow \frac{4000 \text{Å}}{(\lambda_m)_2} = \frac{3500}{2500} = \frac{7}{5}$$

$$\Rightarrow (\lambda_m)_2 = 2857 \text{Å}$$

(NCERT Ed. 2025-26, 11th, Page No. 218)

27. (3)

Speed of the body initially increases and then attains a constant value.

(NCERT Ed. 2025-26, 11th, Page No. 192)

28. (3)

$$\Delta T = \frac{gh}{C_w} = \frac{10 \times 420}{4200} = 1.00^\circ \text{C}.$$

(NCERT Ed. 2025-26, 11th, Page No. 208)

29. (1)

As the area under the curve continuously increases the work done by the system continuously increases.

(NCERT Ed. 2025-26, 11th, Page No. 231)

30. (1)

A is true but R is false.

Viscosity of liquids decreases as temperature rises because thermal agitation reduces cohesive (intermolecular) forces. R claims cohesion increases, which is incorrect.

(NCERT Ed. 2025-26, 11th, Page No. 191)

31. (1)

Heat lost by A = heat gained by B (equal masses):

$$mc_A(40 - 35) = mc_B(35 - 30)$$

$$\Rightarrow 5c_A = 5c_B \Rightarrow c_A : c_B = 1 : 1.$$

(NCERT Ed. 2025-26, 11th, Page No. 209)

32. (4)

$$\text{For monatomic gas: } C_p = \frac{\gamma R}{\gamma - 1} = \frac{(5/3)R}{2/3} = \frac{5R}{2};$$

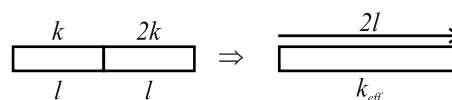
$$C_v = \frac{R}{\gamma - 1} = \frac{R}{2/3} = \frac{3R}{2};$$

$$\text{Adiabatic: } \delta Q = 0 \Rightarrow C = 0;$$

$$\text{Isothermal: } dT = 0 \text{ while } \delta Q \neq 0 \Rightarrow C = \infty$$

(NCERT Ed. 2025-26, 11th, Page No. 253)

33. (4)



Series combination

$$R_{\text{eff}} = R_1 + R_2$$

$$\frac{2l}{k_{\text{eff}} A} = \frac{l}{kA} + \frac{l}{2kA}$$

$$k_{\text{eff}} = \frac{4k}{3}$$

(NCERT Ed. 2025-26, 11th, Page No. 216)

34. (4)

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

$$v_{\text{avg}} = \sqrt{\frac{8RT}{\pi M}}$$

$$v_{\text{mp}} = \sqrt{\frac{2RT}{M}}$$

$$v_{\text{rms}} : v_{\text{mp}} : v_{\text{avg}} = \sqrt{3} : \sqrt{2} : \sqrt{\frac{8}{\pi}}$$

(NCERT Ed. 2025-26, 11th, Page No. 250)

35. (2)

For a soap bubble there are two surfaces, so the work needed to increase area is

$$W = 2T\Delta A.$$

$$W = 2T \cdot 4\pi(R_2^2 - R_1^2) = 8\pi T(0.04 - 0.01)$$

$$= 8\pi(0.5)(0.03) = 0.12\pi \text{ J}.$$

(NCERT Ed. 2025-26, 11th, Page No. 194)

36. (1)

$$\gamma_{(\text{monoatomic})} = \frac{5}{3}$$

$$\gamma_{(\text{diatomic})} = \frac{7}{5}$$

$$\gamma_{(\text{linear triatomic})} = \frac{7}{5}$$

$$\gamma_{(\text{non-linear triatomic})} = \frac{4}{3}$$

(NCERT Ed. 2025-26, 11th, Page No. 253)

37. (2)

Using Pascal's Law

$$\rho_1 g h_1 = \rho_2 g h_2$$

$$\frac{h_1}{h_2} = \frac{\rho_2}{\rho_1}$$

(NCERT Ed. 2025-26, 11th, Page No. 184)

38. (3)

Using Newton's law with average temperature:

First 10 min:

$$\frac{20}{10} = K \left(\frac{70+50}{2} - 30 \right) \Rightarrow 2 = 30K \Rightarrow K = \frac{1}{15}$$

Next 10 min:

$$\frac{50-T}{10} = K \left(\frac{50+T}{2} - 30 \right) = \frac{1}{15} \left(\frac{T-10}{2} \right) = \frac{T-10}{30}$$

$$3(50-T) = T-10$$

$$\Rightarrow 150 - 3T = T - 10 \Rightarrow T = 40^\circ \text{C}$$

(NCERT Ed. 2025-26, 11th, Page No. 220)

39. (4)

$$\eta = \left(1 - \frac{T_2}{T_1} \right) \times 100$$

$$0.60 = 1 - \frac{T_2}{500}$$

$$T_2 = 200\text{K} = -73^\circ \text{C}$$

(NCERT Ed. 2025-26, 11th, Page No. 238)

40. (4)

OA $\rightarrow$ Heating of solid

AB $\rightarrow$ Melting of solid

BC $\rightarrow$ Heating of liquid

CD $\rightarrow$ Boiling of liquid

(NCERT Ed. 2025-26, 11th, Page No. 210)

41. (2)

$$\sigma_1 = \sigma_2$$

Equal areas $\Rightarrow$ same axial stress σ in both rods.

$$Y_1 \alpha_1 \Delta T = Y_2 \alpha_2 \Delta T$$

Thus $Y_1 \alpha_1 = Y_2 \alpha_2$.

(NCERT Ed. 2025-26, 11th, Page No. 207)

42. (4)

Kelvin-planck statement $\rightarrow$ It is impossible to construct a device whose purpose it to convert the entire heat energy absorbed from the source into an equal amount of work.

(NCERT Ed. 2025-26, 11th, Page No. 236)

43. (2)

The stream leaves horizontally, so time of flight to the ground:

$$t = \sqrt{\frac{2h}{g}}$$

For maximum range,

$$h = H/2 \Rightarrow t = \sqrt{H/g} = \sqrt{2.5/10} = 0.5 \text{ s}$$

(NCERT Ed. 2025-26, 11th, Page No. 188)

44. (4)

$$\text{Capillary rise: } h = \frac{2T \cos \theta}{\rho g r}$$

Mass lifted

$$m = \rho \pi r^2 h = \rho \pi r^2 \frac{2T \cos \theta}{\rho g r} = \frac{2\pi T \cos \theta}{g} r \propto r$$

$$\text{So } \frac{m_2}{m_1} = \frac{r_2}{r_1} = \frac{0.9}{0.3} = 3 \Rightarrow m_2 = 3 \times 6 = 18 \text{ g}$$

(NCERT Ed. 2025-26, 11th, Page No. 197)

45. (1)

$$\text{Slope} = \frac{T}{V} = \frac{P}{nR}$$

$$(\text{slope})_{P_1} < (\text{slope})_{P_2}$$

$$P_1 < P_2$$

(NCERT Ed. 2025-26, 11th, Page No. 246)

46. (1)

To force the non-spontaneous direction, an external emf opposing the cell reaction must exceed the cell's own emf; only then does current reverse, giving electrolytic behaviour. Ion flow in the bridge and mere swapping of labels do not achieve this, and a smaller opposing emf cannot overcome the spontaneous tendency.

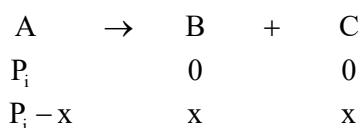
(NCERT Ed. 2025-26 12th Part-I Page No. 32)

47. (4)

Cell potential does not scale with system size, so it is intensive. A negative E° relative to SHE indicates the species prefers oxidation compared to H_2 , hence it is a stronger reducing agent. Faraday's laws relate substance amount to nF , explicitly depending on the reaction's electron stoichiometry, and also state proportionality with charge. Thus all four are correct.

(NCERT Ed. 2025-26 12th Part-I Page No. 40)

48. (1)

Total pressure, $P_t = P_i + x$

$$x = P_t - P_i$$

$$P_i - x = P_i - P_t + P_i = 2P_i - P_t$$

$$K = \frac{2.303}{t} \log \frac{P_i}{2P_i - P_t}$$

(NCERT Ed. 2025-26 12th Part-I Page No. 75)

49. (4)

According to the given graph,

For forward reaction, activation energy is $E_1 + E_2$.

Also, Energy of product > energy of reactant

Energy is inversely related with stability which means products are less stable than reactants.

(NCERT Ed. 2025-26 12th Part-I Page No. 79)

50. (2)

The hydrogen electrode supplies $\frac{1}{2}H_2 \rightleftharpoons H^+ + e^-$,while the silver-silver chloride electrode gives $AgCl(s) + e^- \rightleftharpoons Ag(s) + Cl^-$. Combining these

half-reactions in an HCl medium yields exactly the stated overall reaction; therefore the cell is $Pt | H_2(g) | HCl(aq) || AgCl(s) | Ag(s)$.

(NCERT Ed. 2025-26 12th Part-I Page No. 35)

51. (3)

The balanced overall process is $H_2 + 2Ag^+ \rightarrow 2H^+ + 2Ag$ so $n = 2$ electrons are transferred.

$$\Delta G^\circ = -nFE^\circ$$

$$-2 \times (9.65 \times 10^4) \times 0.50 \text{ J mol}^{-1}$$

$$= -9.65 \times 10^4 \text{ J mol}^{-1}$$

$$= -96.5 \text{ kJ mol}^{-1}$$

(NCERT Ed. 2025-26 12th Part-I Page No. 39)

52. (3)

A stronger reducing agent is more readily oxidised, so the conjugate reduction half-cell must have the most negative E° . Among the listed species, $Fe(s)$ belongs to Fe^{3+}/Fe with $E^\circ = -0.44 \text{ V}$, which is far lower than the positive values for the others; hence $Fe(s)$ is the strongest reducing agent.

(NCERT Ed. 2025-26 12th Part-I Page No. 35)

53. (2)

Since $pH = -\log[H^+]$, multiplying $[H^+]$ by 10^2 changes pH by $-\log(10^2) = -2$. Therefore pH decreases by 2 units.

(NCERT Ed. 2025-26 11th Part-I Page No. 193)

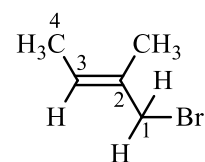
54. (3)

Moles of H^+ from $HCl = 0.500 \times 0.020 = 0.010 \text{ mol}$. From H_2SO_4 , two protons give $0.500 \times 0.010 \times 2 = 0.010 \text{ mol}$. Total $H^+ = 0.020 \text{ mol}$ in 1.000 L

$$\Rightarrow [H^+] = 0.020 \text{ M} \quad \text{Hence } pH = -\log(2 \times 10^{-2}) = 2 - \log 2 = 2 - 0.30 = 1.70$$

(NCERT Ed. 2025-26 11th Part-I Page No. 193)

55. (3)



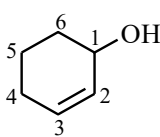
1-Bromo 2-methylbut-2-ene

(NCERT Ed. 2025-26 11th Part-II Page No. 267)

56. (1)

$$\text{CH}_3-\overset{\text{COOH}}{\underset{|}{\text{CH}}}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_3$$

 2-Methylhexanoic acid
 (NCERT Ed. 2025-26 11th Part-II Page No. 267)

57. (4)

 Cyclohex-2-en-1-ol
 (NCERT Ed. 2025-26 11th Part-II Page No. 267)

58. (3)

$$\text{CH}_3\overset{4}{\underset{\text{NH}_2}{\text{CH}}}\overset{3}{\text{CH}}\overset{2}{\text{CH}}\overset{1}{\text{CN}}$$

 3-Aminobutanenitrile
 (NCERT Ed. 2025-26 11th Part-II Page No. 267)

59. (1)

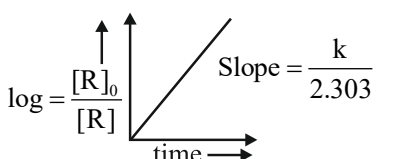
$$\text{HO}-\text{CH}_2-(\text{CH}_2)_3-\text{CH}_2\text{COCH}_3$$

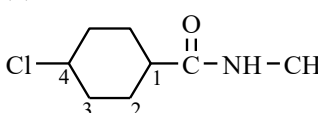
 Correct name is 7-hydroxy heptan-2-one
 (NCERT Ed. 2025-26 11th Part-II Page No. 267)

60. (3)
 An integrated equation of first order reaction is:

$$\log \frac{[\text{R}]_0}{[\text{R}]} = \frac{1}{2.303} kt$$

$$\log \frac{[\text{R}]_0}{[\text{R}]} = \left(\frac{k}{2.303} \right) \times t$$


 (NCERT Ed. 2025-26 12th Part-I Page No. 74)

61. (2)

 4-Chloro-N-methylcyclohexanecarboxamide
 (NCERT Ed. 2025-26 11th Part-II Page No. 267)

62. (2)

$$\text{CH}_3\text{CH}_2\text{COCH}_2\text{CHO}$$
 is condensed formula for 3-oxopentanal
 (NCERT Ed. 2025-26 11th Part-II Page No. 258)

63. (4)

$$\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightarrow 2\text{HI}(\text{g})$$

 Arrhenius equation

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left(\frac{1}{600} - \frac{1}{800} \right)$$

$$\log \frac{1}{2.5 \times 10^{-4}} = \frac{E_a}{2.303 \times 8.314} \left(\frac{200}{600 \times 800} \right)$$

$$\log(2.5 \times 10^{-4}) = \frac{E_a}{2400 \times 2.303 \times 8.314}$$

$$\therefore E_a = 165.5 \text{ kJ/mol}$$

 (NCERT Ed. 2025-26 12th Part-I Page No. 76)

64. (1)
 In presence of acid saponification in pseudo first order.
 (NCERT Ed. 2025-26 12th Part-I Page No. 78)

65. (1)
 - C_nH_{2n} = Alkene
 - $\text{C}_n\text{H}_{2n+2}\text{O}$ = Alcohol
 - $\text{C}_n\text{H}_{2n-2}$ = Alkyne
 - $\text{C}_n\text{H}_{2n}\text{O}$ = Aldehyde
 (NCERT Ed. 2025-26 11th Part-II Page No. 260)

66. (1)

$$\begin{array}{c} \text{CH}_2 - \text{CH} - \text{CH}_2 \\ | \quad | \quad | \\ \text{CN} \quad \text{CN} \quad \text{CN} \end{array}$$

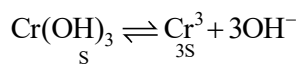
 Propane-1,2,3-tricarbonitrile
 (NCERT Ed. 2025-26 11th Part-II Page No. 267)

67. (2)
 A Lewis base must possess a lone pair for donation. IF_3 , NH_3 and SOCl_2 each have a lone pair on the central atom (I, N and S respectively), enabling donation. In PF_5 the phosphorus is in the PF_5 structure with no lone pair on P, so it cannot donate electron density and thus cannot behave as a Lewis base.
 (NCERT Ed. 2025-26 11th Part-I Page No. 192)

68. (2)
 The ionic product of water K_w increases with temperature because self-ionization is endothermic. Hence both $[\text{H}^+]$ and $[\text{OH}^-]$ rise equally, keeping the solution neutral but lowering $\text{pH} = -\log[\text{H}^+]$. Therefore pH decreases.
 (NCERT Ed. 2025-26 11th Part-I Page No. 194)

69. (4)

Ionisation equation is given as,



$$K_{\text{sp}} = [\text{Cr}^{3+}][\text{OH}^-]^3$$

$$6 \times 10^{-31} = \text{S} \times (3\text{S})^3$$

$$6 \times 10^{-31} = 27 \text{S}^4$$

$$\text{S}^4 = \frac{6 \times 10^{-31}}{27}$$

$$\text{S} = \left(\frac{6}{27} \times 10^{-31} \right)^{1/4} = (2.22 \times 10^{-32})^{1/4}$$

(NCERT Ed. 2025-26 11th Part-I Page No. 204)

70. (1)

$$[\text{H}^+] = [\text{H}^+]_{\text{HNO}_3} + [\text{H}^+]_{\text{H}_2\text{O}} = 10^{-8} + 10^{-7} = 1.1 \times 10^{-7}$$

$$\therefore \text{pH} = -\log(1.1 \times 10^{-7}) = 6.98$$

(NCERT Ed. 2025-26 11th Part-I Page No. 193)

71. (2)

The overall step $\text{Cu}^{2+} \rightarrow \text{Cu}$ equals $\text{Cu}^{2+} \rightarrow \text{Cu}^+$ plus

$$\text{Cu}^+ \rightarrow \text{Cu}, \text{ so } \Delta G_{\text{Cu}^{2+}/\text{Cu}}^\circ = \Delta G_{\text{Cu}^{2+}/\text{Cu}^+}^\circ + \Delta G_{\text{Cu}^+/\text{Cu}}^\circ$$

$$-2\text{FE}_{\text{Cu}^{2+}/\text{Cu}}^\circ = -\text{FE}_{\text{Cu}^{2+}/\text{Cu}^+}^\circ - \text{FE}_{\text{Cu}^+/\text{Cu}}^\circ$$

$$\text{E}_{\text{Cu}^{2+}/\text{Cu}^+}^\circ = 0.32 \times 2 - 0.48 = 0.16 \text{ V}$$

(NCERT Ed. 2025-26 12th Part-I Page No. 40)

72. (1)

For $\text{X} \rightarrow \text{X}^{2+} + 2\text{e}^-$ (anode) and $\text{Y}^{2+} + 2\text{e}^- \rightarrow \text{Y}$ (cathode), $\text{E}_{\text{cell}}^\circ = 0.40 - (-0.40) = 0.80 \text{ V}$. The

$$\text{reaction quotient is } Q = \frac{[\text{X}^{2+}]}{[\text{Y}^{2+}]} = \frac{0.010}{0.10} = 0.10. \text{ Using}$$

$$\text{E} = \text{E}^\circ - \frac{0.06}{2} \log Q \text{ gives } \text{E} = 0.80 - 0.03 \log(0.10)$$

$$= 0.80 - 0.03(-1) = 0.83 \text{ V.}$$

(NCERT Ed. 2025-26 12th Part-I Page No. 38)

73. (2)

At the half-neutralization point, moles of acid converted to acetate equal moles of acid remaining, so $[\text{A}^-] = [\text{HA}]$. By the Henderson-Hasselbalch equation $\text{pH} = \text{pK}_a + \log([\text{A}^-]/[\text{HA}]) = \text{pK}_a$.

Thus $\text{pH} = 4.76$.

(NCERT Ed. 2025-26 11th Part-I Page No. 203)

74. (2)

pOH is largest for the most acidic solution and smallest for the most basic. Strong acid HCl gives the highest pOH, ammonium chloride is acidic by hydrolysis and follows, NaCl is neutral at $\text{pOH} \approx 7$, and strong base NaOH has the lowest pOH. Hence $\text{A} > \text{C} > \text{D} > \text{B}$.

(NCERT Ed. 2025-26 11th Part-I Page No. 203)

75. (2)

Moles of base = $0.025 \times 0.20 = 0.005 \text{ mol}$ and acid = $0.025 \times 0.10 = 0.0025 \text{ mol}$. After neutralization, $0.0025 \text{ mol NH}_4\text{OH}$ remain and $0.0025 \text{ mol NH}_4^+$ are formed in 0.050 L , giving a buffer with $[\text{base}] =$

$$[\text{salt}]. \text{ Using } \text{pOH} = \text{pK}_b + \log\left(\frac{[\text{salt}]}{[\text{base}]}\right)$$

$$= \text{pK}_b = 4.7 \text{ thus } \text{pH} = 14 - 4.7 = 9.3$$

(NCERT Ed. 2025-26 11th Part-I Page No. 203)

76. (2)

NH_4Cl is the salt of a weak base and strong acid; hydrolysis of NH_4^+ makes the solution acidic.

$$\text{K}_h = \frac{\text{K}_w}{\text{K}_b} = \frac{10^{-14}}{10^{-6}} = 10^{-8}. \text{ For concentration}$$

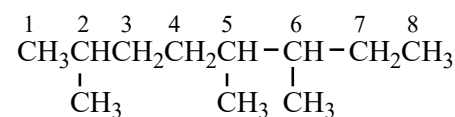
$$\text{C} = 0.040 \text{ M}, [\text{H}^+] = \sqrt{\text{K}_h \text{C}} = \sqrt{10^{-8} \times 0.040} =$$

$$\sqrt{4.0 \times 10^{-10}} = 2.0 \times 10^{-5}.$$

$$\text{Therefore } \text{pH} = 5 - \log 2 = 4.7.$$

(NCERT Ed. 2025-26 11th Part-I Page No. 203)

77. (3)



2,5,6-trimethyloctane

(NCERT Ed. 2025-26 11th Part-II Page No. 267)

78. (2)

Conductance G depends on ionic mobility and number of charge carriers, controlled by the electrolyte's identity, its concentration, and the solvent. The electrode material does not influence the bulk solution conductance measured between them.

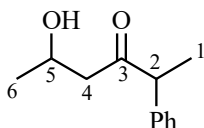
(NCERT Ed. 2025-26 12th Part-I Page No. 41)

79. (3)

On charging the discharge product PbSO_4 reconverts to Pb at the anode and to PbO_2 at the cathode, so A and B are correct. A working lead-acid cell uses spongy Pb as anode and PbO_2 as cathode; statement C mislabels the electrodes when describing the grid, so it is incorrect. Hence A and B only.

(NCERT Ed. 2025-26 12th Part-I Page No. 55)

80. (3)



(NCERT Ed. 2025-26 11th Part-II Page No. 267)

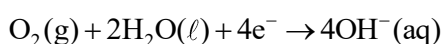
81. (2)

From the Arrhenius form $k = Ae^{-E_a/RT}$, at fixed T a larger E_a yields a smaller k, so (A) is true. The temperature derivative shows $\frac{d(\ln k)}{d(1/T)} = -\frac{E_a}{R}$, so sensitivity increases with E_a , making (B) true. The curvature implies a given ΔT produces a larger fractional change at lower T, so (C) is true. The slope of $\ln k$ vs $1/T$ is $-E_a/R$, not $+E_a/R$; thus (D) is false. Hence three statements are correct.

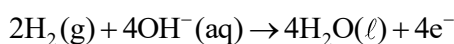
(NCERT Ed. 2025-26 12th Part-I Page No. 79)

82. (3)

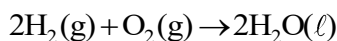
At cathode:



At anode:



Overall reaction:



(NCERT Ed. 2025-26 12th Part-I Page No. 56)

83. (1)

For first order, $t_{1/2} = \frac{0.693}{k} = 231\text{s}$ so (1) is true.

Units differ: rate has concentration time⁻¹ while k has s⁻¹ so (B) is false. A first-order process never reaches zero exactly in finite time, so (C) is false.

(NCERT Ed. 2025-26 12th Part-I Page No. 76)

84. (1)

The cathode is manganese dioxide, where MnO_2 is reduced to a lower oxide (Mn_2O_3); the manganese

oxidation state decreases from +4 to +3. Hence option 1.

(NCERT Ed. 2025-26 12th Part-I Page No. 54)

85. (3)

$$-\frac{1}{3} \frac{d[A]}{dt} = -\frac{1}{2} \frac{d[B]}{dt} = -2 \frac{d[C]}{dt} = \frac{1}{4} \frac{d[P]}{dt}$$

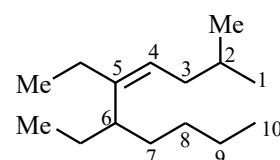
(NCERT Ed. 2025-26 12th Part-I Page No. 65)

86. (2)

The reaction enthalpy ΔH equals $E_{a,\text{fwd}} - E_{a,\text{bwd}} = 150 - 170 = -20 \text{ kJ mol}^{-1}$, a state function independent of path; lowering both barriers equally preserves the difference. Catalysts do not change ΔG° or K; they only change the rate. Hence only 2 is correct.

(NCERT Ed. 2025-26 12th Part-I Page No. 79)

87. (2)



5,6-diethyl-2-methyldec-4-ene

(NCERT Ed. 2025-26 11th Part-II Page No. 267)

88. (1)

The slope has units $\frac{(\Lambda_m)}{\sqrt{c}}$. Since

$[\Lambda_m] = \text{S cm}^2 \text{ mol}^{-1}$ and $[\sqrt{c}] = \text{mol}^{1/2} \text{ L}^{-1/2}$, division yields $\text{S cm}^2 \text{ mol}^{-1} \times \text{mol}^{-1/2} \text{ L}^{1/2} = \text{S cm}^2 \text{ mol}^{-3/2} \text{ L}^{1/2}$.

(NCERT Ed. 2025-26 12th Part-I Page No. 47)

89. (3)

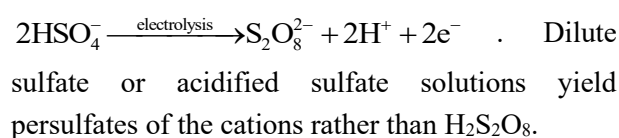
Comparing $1 \rightarrow 2$ keeps $[\text{H}_2]$ constant while doubling $[\text{NO}]$, and the rate becomes four times.

$$\frac{1}{2} = \left(\frac{1}{4}\right)^n \quad n = 2. \text{ Hence rate} \propto [\text{NO}]^2$$

(NCERT Ed. 2025-26 12th Part-I Page No. 77)

90. (2)

Persulfuric acid forms by anodic oxidation of HSO_4^- in cold concentrated sulfuric acid.



(NCERT Ed. 2025-26 12th Part-I Page No. 51)

91. (2)
Diphtheria (Gal ghotu) and leprosy (Kusht rog).
Influenza is a viral disease.
Tobacco Mosaic Virus (TMV) is rod-shaped and appears so when magnified about 1,00,000–1,50,000x.

(NCERT Ed. 2025-26, 12th, Page No. 150)

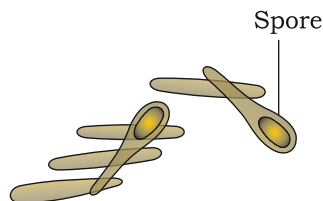
92. (4)
Mycoplasma are the smallest living organisms. Viruses were found to be smaller than bacteria because they passed through bacteria-proof filters. *Saccharomyces* (yeast) is larger than *Mycoplasma*. Viroids are smaller than viruses, and prions are similar in size to viruses.

(NCERT Ed. 2025-26, 11th, Page No. 21)

93. (1)
Bottled fruit juices bought from the market are clearer as compared to those made at home. This is because the bottled juices are clarified by the use of pectinases and proteases.

(NCERT Ed. 2025-26, 12th, Page No. 153)

94. (2)



These bacteria are the rod-shaped Bacilli

(NCERT Ed. 2025-26, 11th, Page No. 12)

95. (1)
Asexual reproduction in fungi is by spores called conidia or sporangiospores or zoospores, and sexual reproduction is by oospores, zygospores, ascospores and basidiospores.

(NCERT Ed. 2025-26, 11th, Page No. 17)

96. (1)
'Toddy', a traditional drink of some parts of southern India is made by fermenting sap from palms.

(NCERT Ed. 2025-26, 12th, Page No. 151)

97. (4)
Lichens are symbiotic associations i.e. mutually useful associations, between algae and fungi. The algal component is known as phycobiont and fungal component as mycobiont, which are autotrophic and heterotrophic, respectively. Algae prepare food for fungi and fungi provide shelter and absorb mineral

nutrients and water for its partner. So close is their association that if one saw a lichen in nature one would never imagine that they had two different organisms within them. Lichens are very good pollution indicators – they do not grow in polluted areas. In the five kingdom classification of Whittaker there is no mention of lichens.

(NCERT Ed. 2025-26, 11th, Page No. 19)

98. (1)
Cyanobacteria are autotrophic microbes widely distributed in aquatic and terrestrial environments many of which can fix atmospheric nitrogen, e.g. *Anabaena*, *Nostoc*, *Oscillatoria*, etc.

(NCERT Ed. 2025-26, 12th, Page No. 158)

99. (2)
Ciliated protozoans have a cavity (gullet) that opens to the outside of the cell surface. The coordinated movement of rows of cilia causes the water laden with food to be steered into the gullet.

(NCERT Ed. 2025-26, 11th, Page No. 16)

100. (2)
This treatment is carried out in two stages:
Primary treatment : These treatment steps basically involve physical removal of particles – large and small – from the sewage through filtration and sedimentation.

The effluent from the primary settling tank is taken for secondary treatment.

Secondary treatment or Biological treatment : The primary effluent is passed into large aeration tanks where it is constantly agitated mechanically and air is pumped into it. This allows vigorous growth of useful aerobic microbes into flocs (masses of bacteria associated with fungal filaments to form mesh like structures). Once the BOD of sewage or waste water is reduced significantly, the effluent is then passed into a settling tank where the bacterial 'flocs' are allowed to sediment. This sediment is called activated sludge. The remaining major part of the sludge is pumped into large tanks called anaerobic sludge digesters.

(NCERT Ed. 2025-26, 12th, Page No. 154)

101. (1)
Protists have a well-defined nucleus and other membrane-bound organelles. The cells of some protists, such as diatoms and dinoflagellates, possess a cell wall, while others, such as protozoans and euglenoids, lack it.

(NCERT Ed. 2025-26, 11th, Page No. 15)

- 102. (1)**
During digestion in anaerobic sludge digester, anaerobic bacteria produce a mixture of gases such as methane, hydrogen sulphide and carbon dioxide.
(NCERT Ed. 2025-26, 12th, Page No. 154)
- 103. (3)**
Sleeping sickness – *Trypanosoma* – DNA
Citrus canker – bacterium – DNA
Potato spindle tuber – Viroid – RNA
Rust in wheat – *Puccinia* (fungus) – DNA
(NCERT Ed. 2025-26, 11th, Page No. 21)
- 104. (1)**
A key belief of the organic farmer is that biodiversity furthers health. The more variety a landscape has, the more sustainable it is. The organic farmer, therefore, works to create a system where the insects that are sometimes called pests are not eradicated, but instead are kept at manageable levels by a complex system of checks and balances within a living and vibrant ecosystem.
(NCERT Ed. 2025-26, 12th, Page No. 156)
- 105. (4)**
Common features between Ascomycetes and Deuteromycetes:
Both have septate and branched mycelium.
Both show heterotrophic mode of nutrition.
Both reproduce asexually by conidia produced exogenously on the special mycelium called conidiophores.
Difference:
Ascomycetes reproduce sexually, forming ascospores endogenously inside an ascus.
Deuteromycetes (Fungi imperfecti) have no known sexual stage.
(NCERT Ed. 2025-26, 11th, Page No. 18)
- 106. (2)**
Products produced using bacteria – dough of idli, streptokinase, butyric acid, and acetic acid.
Products produced using fungi – bread, penicillin, statins, and cyclosporin A.
(NCERT Ed. 2025-26, 12th, Page No. 153)
- 107. (4)**
Bacteria are the most abundant micro-organisms. Bacteria occur almost everywhere. They also live in extreme habitats such as hot springs, deserts, snow and deep oceans where very few other life forms can survive. Bacteria are grouped under four categories based on their shape: the spherical Coccus (pl.: cocci), the rod-shaped Bacillus (pl.: bacilli), the comma-shaped Vibrium (pl.: vibrio) and the spiral Spirillum (pl.: spirilla). Though the bacterial
- structure is very simple, they are very complex in behaviour. Compared to many other organisms, bacteria as a group show the most extensive metabolic diversity.
(NCERT Ed. 2025-26, 11th, Page No. 13)
- 108. (4)**
Slime moulds are saprophytic protists. Under suitable conditions, they form an aggregation called plasmodium which may grow and spread over several feet. During unfavourable conditions, the plasmodium differentiates and forms fruiting bodies bearing spores at their tips. The spores possess true walls. They are extremely resistant and survive for many years, even under adverse conditions. The spores are dispersed by air currents.
(NCERT Ed. 2025-26, 11th, Page No. 15)
- 109. (4)**
Desmid, *Nostoc*, and *Euglena* are photosynthetic as they possess photosynthetic pigments. Although Euglenoids are photosynthetic in the presence of sunlight, when deprived of light they behave like heterotrophs by preying on smaller organisms. *Paramoecium* is heterotrophic.
(NCERT Ed. 2025-26, 11th, Page No. 16)
- 110. (2)**
Streptokinase produced by the bacterium *Streptococcus* and modified by genetic engineering is used as a ‘clot buster’ for removing clots from the blood vessels of patients who have undergone myocardial infarction leading to heart attack.
(NCERT Ed. 2025-26, 12th, Page No. 153)
- 111. (3)**
Aspergillus – fungus – cell wall made of polysaccharides and chitin
Bladderwort – plant – cell wall made of cellulose
Nostoc – cyanobacterium – cell wall made of polysaccharides and amino acids
Desmids – protist – cell wall made of cellulose
(NCERT Ed. 2025-26, 11th, Page No. 11)
- 112. (3)**
Baculoviruses are pathogens that attack insects and other arthropods. The majority of baculoviruses used as biological control agents are in the genus *Nucleopolyhedrovirus*. These viruses are excellent candidates for species-specific, narrow spectrum insecticidal applications. They have been shown to have no negative impacts on plants, mammals, birds, fish or even on non-target insects.
(NCERT Ed. 2025-26, 12th, Page No. 157)
- 113. (4)**
Citrus canker is caused by bacteria.
(NCERT Ed. 2025-26, 11th, Page No. 14)

114. (3)
The Ladybird, and Dragonflies are useful to get rid of aphids and mosquitoes, respectively. An example of microbial biocontrol agents that can be introduced in order to control butterfly caterpillars is the bacteria *Bacillus thuringiensis* (often written as *Bt*). These are available in sachets as dried spores which are mixed with water and sprayed onto vulnerable plants such as brassicas and fruit trees, where these are eaten by the insect larvae. In the gut of the larvae, the toxin is released and the larvae get killed. The bacterial disease will kill the caterpillars, but leave other insects unharmed. Bacteria can fix atmospheric nitrogen while free-living in the soil (examples *Azospirillum* and *Azotobacter*), thus enriching the nitrogen content of the soil.
(NCERT Ed. 2025-26, 12th, Page No. 157)
115. (2)
In two kingdom system of classification, *Spirogyra*, *Chlamydomonas*, *Chlorella*, Fungi, photosynthetic bacteria, mosses, and blue-green algae were all placed under Plantae. *Paramecium* was placed under Animalia.
(NCERT Ed. 2025-26, 11th, Page No. 12)
116. (3)
Saccharomyces → belongs to Ascomycetes (II)
Trichoderma → belongs to Deuteromycetes (IV)
Ustilago → belongs to Basidiomycetes (III)
Trypanosoma → is a Protozoan (I)
(NCERT Ed. 2025-26, 11th, Page No. 18)
117. (2)
M.W. Beijerinck (1898) demonstrated that the extract of the infected plants of tobacco could cause infection in healthy plants and named the new pathogen “virus” and called the fluid as *Contagium vivum fluidum* (infectious living fluid).
(NCERT Ed. 2025-26, 11th, Page No. 20)
118. (3)
After treatment in a sewage treatment plant, most of the organic matter is degraded by microbes, resulting in a decrease in BOD. Therefore, the water released from the sewage treatment plant has a lower BOD compared to the primary effluent.
(NCERT Ed. 2025-26, 12th, Page No. 154)
119. (3)
Kingdom Plantae includes all eukaryotic chlorophyll-containing organisms commonly called plants. A few members are partially heterotrophic such as the insectivorous plants or parasites. Bladderwort and Venus fly trap are examples of insectivorous plants and *Cuscuta* is a parasite.
(NCERT Ed. 2025-26, 11th, Page No. 19)
120. (4)
Microbes are diverse—protozoa, bacteria, fungi and microscopic animal and plant viruses, viroids and also prions that are proteinacious infectious agents.
(NCERT Ed. 2025-26, 12th, Page No. 149)
121. (3)
Puccinia causes rust in wheat. It belongs to Basidiomycetes, which do not possess sex organs. The mycelium is septate and branched, and they generally do not produce asexual spores.
(NCERT Ed. 2025-26, 11th, Page No. 18)
122. (3)
Flocs are masses of bacteria associated with fungal filaments to form mesh like structures that are formed during microbial growth in large aeration tank.
(NCERT Ed. 2025-26, 12th, Page No. 154)
123. (1)
Some hyphae are continuous tubes filled with multinucleated cytoplasm – these are called coenocytic hyphae. It is found in the members of phycmycetes. Members of phycmycetes are found in aquatic habitats and on decaying wood in moist and damp places or as obligate parasites on plants. The mycelium is aseptate and coenocytic. Asexual reproduction takes place by zoospores (motile) or by aplanospores (non-motile). These spores are endogenously produced in sporangium. A zygospore is formed by fusion of two gametes. These gametes are similar in morphology (isogamous) or dissimilar (anisogamous or oogamous). Morels and truffles belong to ascomycetes.
(NCERT Ed. 2025-26, 11th, Page No. 17)
124. (3)
The microbe used for alcohol production is *Saccharomyces cerevisiae* (commonly known as yeast). Yeast is unicellular, not multicellular. Fermentation occurs in anaerobic conditions (absence of oxygen). Yeast is grown in large fermenters for large-scale alcohol production. Yeast belongs to Kingdom Fungi, which includes both unicellular and multicellular forms, not only unicellular.
(NCERT Ed. 2025-26, 12th, Page No. 151)
125. (1)
Archaeobacteria are special since they live in some of the most harsh habitats such as extreme salty areas (halophiles), hot springs (thermoacidophiles) and marshy areas (methanogens). Chemosynthetic autotrophic bacteria play a great role in recycling nutrients like nitrogen, phosphorous, iron and sulphur. Heterotrophic bacteria are most abundant in nature. *Nostoc* colonies are surrounded by gelatinous sheath, mucilaginous sheath.
(NCERT Ed. 2025-26, 11th, Page No. 13)

126. (3)
Micro-organisms such as *Lactobacillus* and others commonly called lactic acid bacteria (LAB) grow in milk and convert it to curd. During growth, the LAB produce acids that coagulate and partially digest the milk proteins.
(NCERT Ed. 2025-26, 12th, Page No. 151)
127. (3)
When a fungus reproduces sexually, two haploid hyphae of compatible mating types come together and fuse. In some fungi the fusion of two haploid cells immediately results in diploid cells (2n). However, in other fungi (ascomycetes and basidiomycetes), an intervening dikaryotic stage (n + n, i.e., two nuclei per cell) occurs; such a condition is called a dikaryon and the phase is called dikaryophase of fungus.
(NCERT Ed. 2025-26, 11th, Page No. 17)
128. (3)
Depending on the type of the raw material used for fermentation and the type of processing (with or without distillation) different types of alcoholic drinks are obtained. Wine and beer are produced without distillation whereas whisky, brandy and rum are produced by distillation of the fermented broth.
(NCERT Ed. 2025-26, 12th, Page No. 152)
129. (1)
Tetanus and cholera are bacterial diseases, and bacteria belong to the kingdom Monera. Smut is caused by a fungus, which belongs to the kingdom Fungi.
Mumps and herpes are viral diseases, and viruses are not included in Whittaker's five-kingdom classification.
(NCERT Ed. 2025-26, 11th, Page No. 19)
130. (3)
Penicillium notatum produces penicillin, the first discovered antibiotic. *Aspergillus niger* is used for the production of citric acid. *Saccharomyces cerevisiae* is also known as baker's and brewer's yeast, as it is used in baking and alcoholic fermentation processes, respectively. *Propionibacterium sharmanii* is used in the production of Swiss cheese.
(NCERT Ed. 2025-26, 12th, Page No. 153)
131. (2)
A virus is a nucleoprotein and the genetic material is infectious. Viruses that infect animals have either single or double stranded RNA or double stranded DNA. Viruses could be crystallised and crystals consist largely of proteins. Bacterial virus usually has double stranded DNA as genetic material.
(NCERT Ed. 2025-26, 11th, Page No. 20)
132. (2)
Statins produced by the yeast *Monascus purpureus* have been commercialised as blood-cholesterol lowering agents. It acts by competitively inhibiting the enzyme responsible for synthesis of cholesterol.
(NCERT Ed. 2025-26, 12th, Page No. 153)
133. (4)
Yeast converts sugars into ethanol and CO₂ during alcoholic fermentation. The large holes in 'Swiss cheese' are due to production of a large amount of CO₂ by a bacterium named *Propionibacterium sharmanii*. Certain bacteria, which grow anaerobically on cellulosic material, produce large amount of methane along with CO₂ and H₂. These bacteria are collectively called methanogens, and one such common bacterium is *Methanobacterium*. *Lactobacillus* converts lactose into lactic acid, which coagulates milk proteins, but no CO₂ is released.
(NCERT Ed. 2025-26, 12th, Page No. 151)
134. (1)
Fungi are also known to form symbiotic associations with plants (mycorrhiza). Many members of the genus *Glomus* form mycorrhiza. The fungal symbiont in these associations absorbs phosphorus from soil and passes it to the plant. Plants having such associations show other benefits also, such as resistance to root-borne pathogens, tolerance to salinity and drought, and an overall increase in plant growth and development. Fungus does not fix atmospheric nitrogen.
(NCERT Ed. 2025-26, 12th, Page No. 158)
135. (2)
Biogas is a mixture of gases (containing predominantly methane) produced by the microbial activity and which may be used as fuel. Bacteria such as *Methanobacterium*, grow anaerobically on cellulosic material and produce large amount of methane along with CO₂ and H₂. Methanogens are present in the rumen (a part of stomach) of cattle.
(NCERT Ed. 2025-26, 12th, Page No. 155)

136. (3)

The cell body and dendrites of the neuron contain Nissl's granules, which are involved in protein synthesis. These granules are not present in the axon or synaptic knob.

(NCERT Ed. 2025-26, 11th, Page No. 232)

137. (2)

Regulation of heartbeat and breathing is controlled by the medulla oblongata (part of hindbrain), not the midbrain.

Integration of visual and auditory signals for orientation is the function of the midbrain (specifically the tectum).

Maintenance of body temperature is regulated by the hypothalamus (part of forebrain).

Coordination of voluntary muscle movement is mainly under the control of the cerebellum and motor cortex.

(NCERT Ed. 2025-26, 11th, Page No. 237)

138. (3)

White muscle fibers (fast-twitch fibers) are adapted for rapid, powerful contractions of short duration. They have less myoglobin (giving them a pale or white appearance), fewer mitochondria, and rely primarily on anaerobic glycolysis for energy. To facilitate rapid contraction and relaxation, they have an extensive and well-developed sarcoplasmic reticulum for quick release and reuptake of Ca^{2+} ions. Therefore, the statement that the amount of sarcoplasmic reticulum is low is incorrect; it is high.

(NCERT Ed. 2025-26, 11th, Page No. 223)

139. (1)

Locomotory actions and changes in body postures are maintained by skeletal muscles whereas transportation of food through digestive tract and transportation of gametes through genital tract are function of smooth muscles, not striated muscles.

(NCERT Ed. 2025-26, 11th, Page No. 218)

140. (1)

List-I	List-II
Tetany	Rapid spasm in skeletal muscle
Osteoporosis	Bone mass decreased
Muscular dystrophy	Progressive degeneration of skeletal muscles
Myasthenia gravis	Auto immune disorder

(NCERT Ed. 2025-26, 11th, Page No. 227)

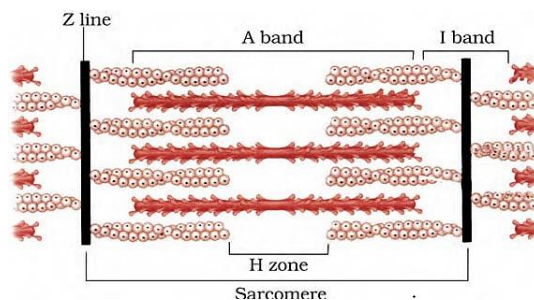
141. (1)

'A' represents A-Band that contains both actin (thin) and myosin (thick) filaments. It does not shorten during muscle contraction.

'B' represents Z-line that marks the boundary of the sarcomere. Actin filaments are anchored to the Z-line.

'C' represents H-Zone which contains only thick filaments (myosin). It reduces during contraction.

'D' represents Actin (thin filament). It shows the thin filament.

(NCERT Ed. 2025-26, 11th, Page No. 220)

142. (4)

Smooth muscle is also known as non-striated muscle because it lacks the regular, repeating arrangement of actin and myosin filaments that create the visible light and dark bands (striations) seen in skeletal and cardiac muscle.

Smooth muscle cells are characteristically spindle-shaped, meaning they are wide in the middle and taper at the ends. This shape is described as fusiform.

(NCERT Ed. 2025-26, 11th, Page No. 218)

143. (2)

The duramater is the outermost meningeal layer. It is described as thick, fibrous and strong.

The piamater is the innermost, delicate, and transparent meningeal membrane. It closely adheres to the brain and spinal cord.

The cranial meninges (duramater, arachnoid, and piamater) are connective tissue membranes.

Arachnoid mater is a key component of the mammalian meninges, which are the protective coverings of the brain and spinal cord.

Cerebrospinal fluid (CSF) provides buoyancy to the brain. It reduces the effective weight of the brain and protects it from mechanical shocks.

(NCERT Ed. 2025-26, 11th, Page No. 235)

144. (2)

The inner parts of cerebral hemispheres and a group of associated deep structures like amygdala, hippocampus, etc., form a complex structure called the limbic lobe or limbic system. Along with the hypothalamus, it is involved in the regulation of sexual behaviour, expression of emotional reactions (e.g., excitement, pleasure, rage and fear), and motivation.

Respiration is an involuntary function controlled by the medulla oblongata in the brainstem.

Body balance is regulated by the cerebellum and vestibular system.

(NCERT Ed. 2025-26, 11th, Page No. 237)

145. (3)

The tibia is the larger and stronger bone of the lower leg. It lies on the medial side (toward the body's midline). It articulates with the femur at the knee joint and with the tarsals at the ankle joint. It bears almost all the body weight.

The fibula is thinner and lateral to the tibia. It does not articulate directly with the femur. It mainly provides muscle attachment and ankle stability, not weight-bearing support.

(NCERT Ed. 2025-26, 11th, Page No. 226)

146. (4)

Locomotion (like walking, swimming, flying) is a type of movement that results in a change of place or position. So, all locomotions are indeed movements.

Movement includes internal motions like heartbeat, peristalsis, blinking, etc., which are not locomotion. Aquatic animals swim, terrestrial animals walk or crawl, aerial animals fly. Locomotion is highly adapted to the habitat.

Apart from searching food or mates, escaping predators is a vital purpose of locomotion.

(NCERT Ed. 2025-26, 11th, Page No. 217)

147. (2)

There are 8 cranial bones in the human skull:

Bone	Number	
Frontal	1	Forehead bone
Parietal	2	on the sides and top of skull
Temporal	2	Near the ears
Occipital	1	Back and base of the skull
Sphenoid	1	Bat-shaped bone; floor of skull
Ethmoid	1	Between eyes

(NCERT Ed. 2025-26, 11th, Page No. 224)

148. (3)

LMM (Light Meromyosin) is the tail portion of the myosin molecule, which does not have ATP binding sites.

ATP binding sites are found in HMM (Heavy Meromyosin), particularly on the globular head of myosin.

(NCERT Ed. 2025-26, 11th, Page No. 221)

149. (4)

Ciliary movement involves microtubules arranged in a 9+2 arrangement and driven by dynein arms.

Muscular movement involves sarcomeres, the contractile units of skeletal muscles, made up of actin and myosin filaments.

(NCERT Ed. 2025-26, 11th, Page No. 218)

150. (3)

The potential difference across the membrane of neuron at rest is called resting membrane potential. The Na^+/K^+ pump (an active transport mechanism) does play a major role in maintaining resting potential.

However, passive mechanisms such as leakage channels (especially for K^+ ions) also contribute significantly to maintaining the resting potential.

(NCERT Ed. 2025-26, 11th, Page No. 232)

151. (2)

Upon a single stimulation Ca^{2+} is released from the sarcoplasmic reticulum i.e. skeletal muscle contraction can occur.

However, Ca^{2+} cannot be pumped back efficiently into the sarcoplasmic reticulum that's why why cytoplasmic Ca^{2+} remains high. This leads to delayed relaxation since troponin stays activated longer.

(NCERT Ed. 2025-26, 11th, Page No. 223)

152. (3)

Cardiac muscle cells have an abundant number of mitochondria, enabling continuous aerobic respiration, which provides a sustained energy supply.

Dense capillary networks ensure constant oxygen and nutrient delivery.

Together, these features directly contribute to fatigue resistance.

(NCERT Ed. 2025-26, 11th, Page No. 223)

153. (3)

List-I	List-II
Sympathetic nerve	Mobilises body system during emergency situation
Para sympathetic nerve	Promotes non emergency function
Axon	Nerve impulse travel in centrifugal manner
Dendrites	Nerve impulse travel in centripetal manner

(NCERT Ed. 2025-26, 11th, Page No. 232)

154. (4)

Atlanto-occipital joint, a condyloid joint allows flexion and extension (the "yes" motion).

Atlanto-axial joint, a pivot joint allows rotation (the "no" motion).

The elbow is a classic example of a hinge joint.

The pubic symphysis is a cartilaginous joint (specifically, a symphysis), not a synovial joint. It allows only slight movement, not a wide range of motion.

Sutures are immovable fibrous joints found between the bones of the skull.

(NCERT Ed. 2025-26, 11th, Page No. 227)

155. (3)

The cerebellum is mainly responsible for coordination of voluntary movements, posture, balance, and motor learning.

It does not process sensations like touch and pain.

Those functions are handled by the somatosensory cortex in the parietal lobe of the cerebrum.

(NCERT Ed. 2025-26, 11th, Page No. 236)

156. (4)

The occipital bone is at the posterior (not anterior) part of the skull. It does not articulate with the nasal bones. The pituitary fossa (sella turcica) is housed in the sphenoid bone.

(NCERT Ed. 2025-26, 11th, Page No. 224)

157. (3)

Lower invertebrates (e.g., sponges, cnidarians) have very simple or no true neural organisation. Complexity increases in higher invertebrates like arthropods and annelids.

Neurons are specialised to detect, receive, and transmit stimuli via electrical impulses. Transmission is a primary function of neurons.

(NCERT Ed. 2025-26, 11th, Page No. 231)

158. (3)

Muscle tissue (skeletal, cardiac, smooth) is derived from the mesoderm. Ectoderm gives rise to the nervous system and epidermis.

Muscle tissue makes up about 40–50% of body weight in healthy adults, especially in males.

Skeletal muscles produce heat during contraction (e.g., shivering), contributing significantly to thermoregulation. This is called thermogenesis.

Only skeletal muscle is under voluntary control. Cardiac and smooth muscles (e.g., in heart, intestines, blood vessels) contract involuntarily.

(NCERT Ed. 2025-26, 11th, Page No. 218)

159. (2)

Depolarisation occurs when Na⁺ channels open, allowing sodium ions to rush into the neuron.

The inner surface becomes temporarily more positive than the outside.

Membrane potential may rise to around +30 mV.

This is the correct phase where the inside becomes positively charged relative to the outside.

(NCERT Ed. 2025-26, 11th, Page No. 233)

160. (4)

Tropomyosin is a rod-shaped protein that lies along actin filaments, blocking myosin-binding sites. It does not bind calcium.

Troponin-C (TnC) is the actual calcium-binding protein that initiates contraction by moving tropomyosin aside.

(NCERT Ed. 2025-26, 11th, Page No. 222)

161. (1)

List-I	List-II
Amoeboid movement	Phagocytic action of a neutrophil
Ciliary movement	Passage of ova through the female reproductive tract
Muscular movement	Contraction of the biceps muscle
Protoplasmic Streaming	Movement within the cell of a plant like <i>Hydrilla</i>

(NCERT Ed. 2025-26, 11th, Page No. 217)

162. (2)

The cerebellum is responsible for coordination, precision, and balance of movements.

It does not initiate voluntary muscle activity.

Voluntary motor commands originate from the primary motor cortex in the frontal lobe of the cerebrum.

(NCERT Ed. 2025-26, 11th, Page No. 236)

163. (1)

Bone is classified as a specialized connective tissue because it supports and protects body structures and has a matrix and cells like typical connective tissues, but is highly adapted for strength and rigidity.

The hardness of bone comes from inorganic salts like calcium phosphate, while collagen fibres provide tensile strength and flexibility. This composition gives bone its characteristic strength and rigidity.

(NCERT Ed. 2025-26, 11th, Page No. 224)

164. (1)

The correct functional order for a motor command originating in the brain to reach a gland or visceral organ is:

C. Central Nervous System (CNS): The command originates in the brain, which, along with the spinal cord, constitutes the CNS. This is the starting point for the motor impulse.

B. Peripheral Nervous System (PNS): For the signal to travel from the CNS to any other part of the body, it must pass through the nerves of the PNS. The PNS encompasses all the nerves outside of the brain and spinal cord, acting as the communication network.

A. Autonomic Nervous System (ANS): Since the target is a gland or a visceral organ, the control is involuntary. The Autonomic Nervous System is the specific functional division of the PNS that governs these involuntary actions.

D. Parasympathetic Division: The ANS is further divided into the sympathetic and parasympathetic divisions. The parasympathetic division is one of these two branches that directly sends signals to visceral organs and glands to regulate their functions (often referred to as "rest and digest" responses).

(NCERT Ed. 2025-26, 11th, Page No. 231)

165. (4)

The Atlas is not a thoracic vertebra. It is actually the first cervical vertebra (C₁). It supports the skull. Thoracic vertebrae are in the mid-back and support the ribs, starting from T₁.

The acetabulum is not part of the femur. It is a cup-shaped socket in the hip bone (pelvis). It articulates with the head of the femur (not humerus). The humerus fits into the glenoid cavity of the scapula.

(NCERT Ed. 2025-26, 11th, Page No. 225)

166. (2)

The part of the Central Nervous System (CNS) that coordinates and maintains rhythmic activity as the body's master clock is the hypothalamus (Suprachiasmatic nucleus).

The suprachiasmatic nucleus (SCN), a tiny cluster of about 20,000 nerve cells located in the hypothalamus, serves as the primary regulator of the body's circadian rhythms. This internal "master clock" generates an approximately 24-hour cycle that influences a wide range of physiological processes.

(NCERT Ed. 2025-26, 11th, Page No. 236)

167. (4)

The femur (thigh bone) is the longest and strongest bone in the human body. It supports the entire upper body's weight when standing.

While sitting, the femur does not bear the full weight of the upper body. Instead, the ischial tuberosities of the pelvic bones primarily bear the load.

(NCERT Ed. 2025-26, 11th, Page No. 226)

168. (2)

Electrical synapses do not have a synaptic cleft; instead, pre- and post-synaptic neurons are in close contact via gap junctions, allowing direct ionic current flow.

In electrical synapses, ionic current passes directly between neurons—just like conduction through an axon, making it fast and bidirectional, without synaptic delay.

Neurotransmitters (e.g., acetylcholine, GABA, dopamine) bind to receptors and generate excitatory postsynaptic potential (EPSP) or inhibitory postsynaptic potential (IPSP).

Presynaptic membrane (from axon terminal) and postsynaptic membrane (usually on dendrite or soma) are involved in signal transmission.

(NCERT Ed. 2025-26, 11th, Page No. 234)

169. (1)

Impulse flows in the following steps:-

(i) Receptor generates an impulse.
(ii) Impulse travels through dendrite (D) of sensory neuron.

(iii) Then to the cell body (CB) of the sensory neuron.

(iv) Then through its axon (A) to reach the synapse (S).

(v) Impulse is transmitted across the synapse to the dendrite (D) of the motor neuron.

(vi) Then to cell body of the motor neuron.

(vii) Then along its axon (A) to the effector (muscle/gland).

(NCERT Ed. 2025-26, 11th, Page No. 232)

170. (2)

Bones in One Hand:

Carpals (wrist bones) = 8

Metacarpals (palm bones) = 5

Phalanges (finger bones) = 14

Total per hand = $8 + 5 + 14 = 27$

Total for both hands = $27 \times 2 = 54$

Bones in One Foot:

Tarsals (ankle bones) = 7

Metatarsals (foot arch bones) = 5

Phalanges (toe bones) = 14

Total per foot = $7 + 5 + 14 = 26$

Total for both feet = $26 \times 2 = 52$

While the human skeleton has 206 bones, 106 (54 + 52) of them are concentrated in just the hands and feet

(NCERT Ed. 2025-26, 11th, Page No. 226)

171. (1)

Ion channels are integral membrane proteins. They facilitate passive diffusion of specific ions (like Na^+ , K^+ , Cl^- , Ca^{2+}) along their electrochemical gradients. No ATP is needed, this is passive transport. Vital in action potential generation and propagation in neurons.

A core feature of the nervous system is sensitivity/irritability i.e the ability to detect stimuli and respond (e.g., generating action potentials and reflexes).

(NCERT Ed. 2025-26, 11th, Page No. 232)

172. (3)

List-I	List-II
Sternum	Flat Bone
Humerus	Long Bone
Vertebra	Irregular Bone
Carpal	Short Bone

(NCERT Ed. 2025-26, 11th, Page No. 225)

173. (2)

Our vertebral column is formed by 33 serially arranged units called vertebrae in early life (before fusion).

The vertebral column is dorsally placed (i.e., toward the back). It forms the posterior axis of the body and protects the spinal cord.

The first vertebra is the atlas (C1). The atlas articulates with the occipital condyles of the skull. The second vertebra is the axis (C2), which forms the pivot joint with the atlas.

(NCERT Ed. 2025-26, 11th, Page No. 225)

174. (3)

Osteoporosis is characterized by low bone mass and structural deterioration of bone tissue, leading to increased fragility. Estrogen plays a critical role in maintaining bone homeostasis by regulating the activity of bone-resorbing and bone-forming cells. Estrogen normally suppresses osteoclast differentiation and survival.

(NCERT Ed. 2025-26, 11th, Page No. 227)

175. (1)

List-I	List-II
Corpora quadrigemina	Four round swellings in the midbrain
Spinal cord	Attached to medulla
Cerebral aqueduct	Canal passing through the midbrain
Limbic system	The inner parts of cerebral hemispheres, amygdala and hippocampus

(NCERT Ed. 2025-26, 11th, Page No. 236)

176. (3)

The 11th and 12th rib pairs are called floating ribs because they do not attach to the sternum either directly or indirectly. They are only attached posteriorly to the vertebral column.

They're not "floating" to allow greater diaphragm movement. Their naming is due to lack of anterior attachment; they mainly provide protection (e.g., kidneys) and muscle attachment.

(NCERT Ed. 2025-26, 11th, Page No. 225)

177. (3)

Synovial joints have a synovial cavity filled with synovial fluid, which reduces friction and lubricates the articulating surfaces.

Synovial joints are designed for free movement, they are the most mobile joints in the body. Minimal movement is a feature of fibrous or cartilaginous joints.

Thumb joint (between metacarpal and trapezium) is a saddle joint — a type of synovial joint. Knee joint is a hinge-type synovial joint.

Ball and socket joints (like shoulder and hip) are a type of synovial joint, allowing multi-axial movement.

(NCERT Ed. 2025-26, 11th, Page No. 227)

178. (1)

A - Sarcomere is a structural unit, while all are proteins.

B - Tropomyosin is a protein, while the rest are structural muscle components.

C - Myosin is a motor protein, while the rest are regulatory/structural proteins.

D - The "ATPase enzyme" function, the "ATP binding site," and the "Actin binding site" are all located on the head of the myosin molecule. The "Myosin binding site," however, is the location on the actin molecule where the myosin head attaches.

(NCERT Ed. 2025-26, 11th, Page No. 224)

179. (3)

Association areas are distributed in the frontal, parietal, temporal, and occipital lobes of the cerebral cortex.

They link sensory inputs with stored experiences, enabling perception, memory, intelligence, language, and decision-making.

Unlike primary sensory or motor areas, they do not directly receive sensory input or generate motor output — instead, they interpret, integrate, and plan.

(NCERT Ed. 2025-26, 11th, Page No. 236)

**180. (2)**

List-I	List-II
Voluntary muscle fibre	Cylindrical, multinucleated, striated
Cardiac muscle fibre	Branched, uninucleated, with intercalated discs
Visceral muscle fibre	Spindle-shaped, uninucleated, involuntary
Fascicle	Bundles of skeletal muscle fibres wrapped in perimysium

(NCERT Ed. 2025-26, 11th, Page No. 219)



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